LAN CONFIGURATION AND NETWORK SERVICES ASSIGNMENT

Static addressing • DHCP • DNS • VLANs • NAT

1. Assign Static IP Addresses to Five Devices and Verify Connectivity

Objective: Configure five computers with manually assigned IP addresses in one subnet and verify that they can communicate using ping.

Sample network plan

Device	IPv4 address	Subnet mask	Default gateway
PC-1	192.168.1.11	255.255.255.0	192.168.1.1
PC-2	192.168.1.12	255.255.255.0	192.168.1.1
PC-3	192.168.1.13	255.255.255.0	192.168.1.1
PC-4	192.168.1.14	255.255.255.0	192.168.1.1
PC-5	192.168.1.15	255.255.255.0	192.168.1.1

All five devices belong to the 192.168.1.0/24 network. Each device has a unique host address. The gateway shown is an example; use the actual router address in a real network. Ensure these static addresses are outside the router's DHCP pool or reserved to avoid address conflicts.

Configuration procedure

1. Connect all five devices to the same switch or the same LAN/Wi-Fi network.
2. On each Windows device, open Settings > Network & Internet > Advanced network settings > More network adapter options. Open the active adapter's Properties.
3. Select Internet Protocol Version 4 (TCP/IPv4) > Properties > Use the following IP address.
4. Enter the address, subnet mask and gateway from the table. Enter a valid DNS server if name resolution is also required. For this local ping test, DNS is not required.
5. Repeat the configuration for all five devices, ensuring that no two devices use the same IP address.
6. Open Command Prompt on PC-1 and run ipconfig to confirm its address. Repeat or verify on the remaining computers.
7. From PC-1, ping each of the other four addresses. Then test at least one reverse-direction ping from another device.

Connectivity test commands

```
ipconfig
ping 192.168.1.12
ping 192.168.1.13
ping 192.168.1.14
ping 192.168.1.15
```

Expected result: Successful replies show that the destination device is reachable. A message such as “Request timed out” may indicate that the device is offline, the address is incorrect, the devices are on different networks/VLANs, or a firewall is blocking ICMP. Confirm the IP settings and allow ICMP echo requests for the lab if necessary.

Conclusion

Static addressing gives each device a predictable address. Correct subnet masks and unique IP addresses allow devices in the same subnet to communicate directly without needing a router.

2. Configure DHCP to Assign Addresses to at Least Three Devices

Objective: Configure a DHCP server so that three or more clients automatically receive unique IP addresses from a defined pool.

Example topology and addressing

```
DHCP Server / Router  ----  Switch  ----  PC-1, PC-2, PC-3
LAN network: 192.168.20.0/24
Router / default gateway: 192.168.20.1
DHCP pool: 192.168.20.100 - 192.168.20.150
```

Router DHCP example (Cisco IOS)

```
enable
configure terminal
interface gigabitEthernet 0/0
 ip address 192.168.20.1 255.255.255.0
 no shutdown
exit
ip dhcp excluded-address 192.168.20.1 192.168.20.99
ip dhcp pool LAN-POOL
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 8.8.8.8
exit
end
show ip dhcp pool
show ip dhcp binding
```

The commands are a sample for a Cisco IOS router. Interface names vary by model and simulator. If using Packet Tracer's Server device instead, open Services > DHCP, turn the service On, and enter the pool name, gateway, DNS server, starting IP address, subnet mask and maximum number of users.

Client configuration and verification

8. Connect at least three PCs to the same LAN or VLAN served by this DHCP pool.
9. On each Windows PC, open the IPv4 adapter properties and select Obtain an IP address automatically and Obtain DNS server address automatically.
10. Run ipconfig /release followed by ipconfig /renew to request a fresh lease.
11. Run ipconfig /all on each client and record the IPv4 address, subnet mask, DHCP server, default gateway and lease information.
12. On the router, run show ip dhcp binding to check the addresses leased to clients.

Client	Expected example lease	Verification
PC-1	192.168.20.100	Unique address in DHCP pool
PC-2	192.168.20.101	Different from PC-1
PC-3	192.168.20.102	Different from PC-1 and PC-2

Actual leases may be assigned in a different order. The important checks are that each client receives an address in the intended range, all addresses are unique, and the gateway and subnet mask are correct.

Conclusion

DHCP reduces manual configuration and helps prevent typing errors and duplicate IP addresses. The server maintains leases and supplies clients with essential settings such as the subnet mask, default gateway and DNS server.

3. Configure DNS for a Custom Local Hostname

Objective: Make a hostname such as musicplayer.local resolve to the IP address of a local web server or device interface, so that a connected client can open it in a browser.

Example: A local web server has the address 192.168.20.10 and hosts a test page. The hostname musicplayer.local will be mapped to that address.

Option A: Configure a local DNS service

13. Ensure the web server has a stable IP address or a DHCP reservation. In this example, use 192.168.20.10/24.
14. On a Linux lab machine, install dnsmasq using the package manager available for the distribution.
15. Add a local hostname mapping in the dnsmasq configuration. A typical entry is address=/musicplayer.local/192.168.20.10.
16. Restart the dnsmasq service and confirm that it is running.
17. Configure the router's DHCP settings to advertise the IP address of the dnsmasq machine as the DNS server. Renew client leases so they receive the updated DNS setting.
18. From a client, run nslookup musicplayer.local and confirm that it returns 192.168.20.10.
19. Open http://musicplayer.local in the browser. Confirm that the local web page or device interface loads.

Example dnsmasq mapping

```
address=/musicplayer.local/192.168.20.10
```

The exact configuration-file location and service commands depend on the Linux distribution. Make sure the DNS service listens on the LAN interface and that firewall rules allow DNS requests on UDP/TCP port 53.

Option B: Quick hosts-file test on Windows

20. Open Notepad as Administrator.
21. Open C:\Windows\System32\drivers\etc\hosts and change the file filter to All files.
22. Add the line shown below and save the file.
23. Run ipconfig /flushdns, then run ping musicplayer.local or nslookup is not suitable for verifying a hosts-file-only entry; use ping or another normal resolver-aware application.
24. Open http://musicplayer.local in a browser. The local web server must be running at the mapped IP address.

```
192.168.20.10    musicplayer.local
```

A hosts-file entry applies only to the computer where it is added. It does not configure DNS for every device. For network-wide access, use a local DNS service and distribute its address through DHCP. Also note that .local is commonly used by multicast DNS/Bonjour; for a controlled lab, a reserved subdomain such as musicplayer.home.arpa may avoid conflicts with multicast name resolution.

Verification and troubleshooting

```
ipconfig /all
nslookup musicplayer.local
ping musicplayer.local
```

If the hostname does not resolve, check the DNS server configured on the client, the record syntax, DNS service status, firewall rules and client cache. If the name resolves but the page does not open, test the IP directly in the browser and confirm that the web service is running and listening on the expected port.

4. Create Two VLANs and Test Communication

Objective: Divide a switch network into two separate Layer 2 broadcast domains. Devices in the same VLAN should communicate, while devices in different VLANs should not communicate when no inter-VLAN routing is configured.

Topology and address plan

```
                Managed Switch
        PC-1 ---- |             | ---- PC-3
        PC-2 ---- |             | ---- PC-4
  VLAN 10: Workstations      VLAN 20: Smart Devices
```

Device	VLAN	Switch access port	Sample IP / mask
PC-1	10 – Workstations	Fa0/1	192.168.10.11 /24
PC-2	10 – Workstations	Fa0/2	192.168.10.12 /24
Device-3	20 – Smart Devices	Fa0/3	192.168.20.11 /24
Device-4	20 – Smart Devices	Fa0/4	192.168.20.12 /24

Use a managed switch or a simulator such as Cisco Packet Tracer. Configure static IP addresses for this demonstration. Do not connect a router or Layer 3 switch to route between the VLANs during the isolation test.

Cisco IOS switch configuration

```
enable
configure terminal
vlan 10
 name Workstations
exit
vlan 20
 name Smart_Devices
exit
interface range fastEthernet 0/1-2
 switchport mode access
 switchport access vlan 10
exit
interface range fastEthernet 0/3-4
 switchport mode access
 switchport access vlan 20
exit
end
show vlan brief
```

Port numbering varies by switch model. If the switch requires VLAN creation or access-port configuration through a graphical interface, create VLAN IDs 10 and 20, assign the indicated ports, and save the configuration.

Connectivity tests

25. From PC-1, ping PC-2 at 192.168.10.12. This should succeed because both devices are in VLAN 10 and the same IP subnet.
26. From Device-3, ping Device-4 at 192.168.20.12. This should succeed because both devices are in VLAN 20 and the same IP subnet.
27. From PC-1, ping Device-3 at 192.168.20.11. This should fail when there is no inter-VLAN routing.
28. Check the switch using show vlan brief and verify that ports Fa0/1–Fa0/2 belong to VLAN 10 and Fa0/3–Fa0/4 belong to VLAN 20.

Expected test results

Source	Destination	Expected result	Reason
PC-1 (VLAN 10)	PC-2 (VLAN 10)	Success	Same VLAN and subnet
Device-3 (VLAN 20)	Device-4 (VLAN 20)	Success	Same VLAN and subnet
PC-1 (VLAN 10)	Device-3 (VLAN 20)	Failure	No inter-VLAN routing configured

If same-VLAN pings fail, check IP addresses, subnet masks, switch-port assignments, cable/link status and host firewalls. If different-VLAN pings succeed, check whether a router, Layer 3 switch or other path is providing inter-VLAN routing. VLAN separation alone does not prevent communication if routing and suitable policies are configured.

5. Simulate Internet Access Using NAT

Objective: Configure a router to translate private LAN addresses into an outside address so that internal devices can communicate with an external network or a simulated external server.

Example topology and address plan

```
LAN PCs ---- Switch ---- Router (inside) ---- Router (outside) ---- ISP / external test server
Inside LAN: 192.168.30.0/24
Router inside interface: 192.168.30.1/24
Router outside interface: 203.0.113.2/30 (lab documentation example)
Next-hop ISP router: 203.0.113.1
External test server: 198.51.100.10 (lab documentation example)
```

The addresses 203.0.113.0/24 and 198.51.100.0/24 are reserved for documentation. They are examples for a closed simulation and are not real public Internet destinations. In Packet Tracer, create an ISP router and an external server using a consistent simulated addressing plan.

Cisco IOS PAT/NAT overload example

```
enable
configure terminal
interface gigabitEthernet 0/0
 description LAN
 ip address 192.168.30.1 255.255.255.0
 ip nat inside
 no shutdown
exit
interface gigabitEthernet 0/1
 description WAN
 ip address 203.0.113.2 255.255.255.252
 ip nat outside
 no shutdown
exit
access-list 1 permit 192.168.30.0 0.0.0.255
ip nat inside source list 1 interface gigabitEthernet 0/1 overload
ip route 0.0.0.0 0.0.0.0 203.0.113.1
end
show ip nat translations
show ip nat statistics
```

Interface names and command support vary by router. Configure the ISP-side router with a return route to the LAN through 203.0.113.2, or use a suitable simulated upstream network. Without a return path, replies may not reach the translated client.

Client setup and testing

29. Configure a LAN PC with an address such as 192.168.30.100/24, default gateway 192.168.30.1 and a working DNS server. Alternatively, use DHCP to supply these settings.
30. Ping the LAN gateway 192.168.30.1 to verify local connectivity.
31. Ping the ISP next-hop address 203.0.113.1, then the simulated external server 198.51.100.10. These tests verify routing independently of DNS.
32. On the router, run show ip nat translations while the client generates traffic. Confirm that the inside local address (for example 192.168.30.100) is translated to the outside interface address.
33. If the simulated external server hosts a web page, open its IP address in the client browser. For a domain-name test, configure DNS to resolve the test hostname to the external server address.
34. Test a real external site such as www.wikipedia.org only when the simulator has an actual Internet/upstream connection and working DNS. Packet Tracer's isolated simulation does not automatically provide real Internet access.

How NAT works

NAT overload, also called Port Address Translation (PAT), allows multiple private LAN devices to share one outside IPv4 address. The router tracks each connection using address and port information, translates outgoing packets, and maps returning traffic back to the correct internal device.

Common NAT troubleshooting checks

- Confirm the LAN PC has the correct IP address, subnet mask, gateway and DNS configuration.
- Verify that the LAN-facing interface is marked ip nat inside and the WAN-facing interface is marked ip nat outside.
- Check that the NAT access list matches the correct private subnet.
- Confirm the default route points to the ISP next hop and that the upstream router has a return route.
- Check interface status, ACL/firewall rules and NAT translation counters.
- Test by IP first; troubleshoot DNS separately if IP connectivity works but domain names fail.

Practical Evidence Checklist and Overall Conclusion

Capture evidence from your own lab or simulator for each activity. The following screenshots make the work easier to verify:

- Static IP addressing: show the five-device topology or IP configuration and successful ping replies between devices.
- DHCP: show the DHCP pool configuration, three client ipconfig /all results, and the router's show ip dhcp binding output where applicable.
- DNS: show the local DNS record or hosts-file entry, nslookup result, and the local page opened using the custom hostname.
- VLANs: show the switch VLAN table (show vlan brief) and ping results for same-VLAN success and cross-VLAN failure.
- NAT: show the topology, router NAT configuration, successful test to the external/simulated server, and show ip nat translations output.

Conclusion

This assignment demonstrates five essential LAN administration tasks. Static IP configuration provides predictable addressing, DHCP automates client configuration, DNS maps names to addresses, VLANs logically separate devices into different broadcast domains, and NAT allows private IPv4 networks to communicate with an upstream network. Each feature should be verified with configuration commands and connectivity tests rather than assuming that a saved configuration is working.

Important note: All IP addresses, interface names, topologies and command examples are illustrative. Adapt them to the devices and simulator being used. A ping failure does not always prove a network fault because host firewalls may block ICMP. Record actual output from your own lab rather than presenting sample values as captured results.